

Project Title Goes Here

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Abstract

Write a 150–250 word summary of your research question, methods, and key findings here.

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1 Introduction

Understanding the effect of a treatment, intervention, or design decision often requires more than a single observation — it requires a controlled experiment analyzed with sound statistical methods. In this project, we investigate *[state your research question here, e.g., "whether sentiment expressed in commit messages is associated with the likelihood that a commit introduces a bug"]*.

This question matters because *[briefly justify why the problem is worth studying — practical impact, gap in prior work, or a decision that depends on the answer]*. Prior studies such as [Smith and Doe \[2020\]](#) have examined related phenomena, but *[state the specific gap or limitation your project addresses]*.

The remainder of this report is organized as follows. Section 2 reviews relevant prior work. Section 3 describes the dataset and statistical methods used. Section 4 reports the statistical results, which we interpret and discuss in Section 5. Finally, Section 6 summarizes our contributions and outlines directions for future work.

2 Related Works

Several studies have examined related questions using statistical or experimental methods. [Smith and Doe \[2020\]](#) studied *[brief description of their setup]* and found that *[their key finding]*. Similarly, [\[Jones and Brown, 2019\]](#) reported *[finding]*, using a *[method, e.g., randomized controlled trial / observational study]* design.

Other work has taken a different methodological approach. For example, [Lee \[2021\]](#) relied on *[method]* rather than *[alternative method]*, which *[state the trade-off or limitation this introduces]*.

Taken together, this body of work suggests *[synthesize the common thread across the cited studies]*. However, *[identify what remains unanswered or contested]* — a gap this project aims to address.

3 Methodology

3.1 Dataset

We use a dataset consisting of *[N observations/records]* collected from *[source, e.g., a public repository, a survey, a lab experiment]*. Each observation includes the following variables:

- **[Variable 1]** — *[type: continuous / categorical / binary; brief description]*
- **[Variable 2]** — *[type; description]*
- **[Outcome variable]** — *[type; description]*

Raw data is stored under `data/`; any cleaning or preprocessing steps (e.g., removing missing values, outlier filtering, normalization) should be scripted under `code/` so the pipeline is reproducible rather than performed manually.

3.2 Statistical Methods

To test our hypotheses, we use [*e.g., a two-sample t -test / one-way ANOVA / linear regression / chi-squared test*]. We report effect sizes (e.g., Cohen’s d) alongside p -values, and we set the significance threshold at $\alpha = 0.05$ unless stated otherwise. Where multiple comparisons are performed, we apply [*e.g., a Bonferroni correction*] to control the family-wise error rate.

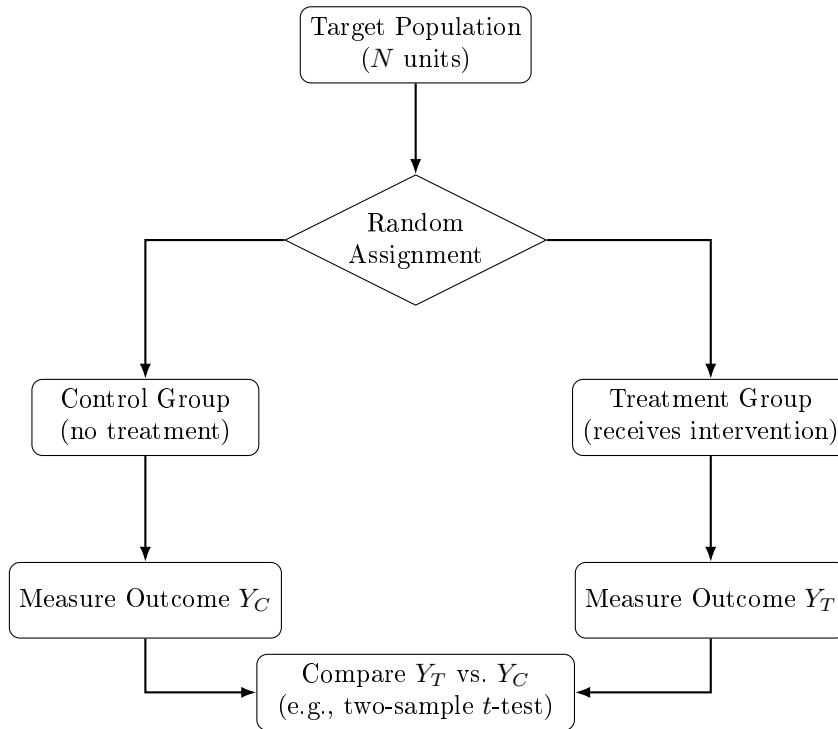


Figure 1: Randomized two-group experimental design.

All analysis code is written in *[Python]* and is available in `code/`, along with instructions for reproducing the results in Section 4.

4 Results

Table 1 in Appendix A illustrates the table format we use to report results; replace it with your actual findings here (or move a copy of the example table code into this section and update the numbers).

[*Describe your key result in words first, then support it with statistics, e.g.:*] The proposed group showed a higher mean outcome ($M = [value]$, $SD = [value]$) than the baseline group ($M = [value]$, $SD = [value]$). This difference was [*statistically significant / not statistically significant*], $t([df]) = [t-value]$, $p = [p-value]$, with an effect size of $d = [value]$.

Figure 2 in Appendix A shows the corresponding figure format; replace the placeholder plot with your own visualization (e.g., a boxplot of the two groups, a scatter plot with a fitted regression line, or a bar chart of proportions).

If you ran multiple tests, summarize them together, e.g., in a single table with one row per hypothesis/comparison, including the test statistic, degrees of freedom, p -value, and effect size for each.

[Report results neutrally here — save interpretation, caveats, and comparison with prior work for Section 5.]

5 Discussion

5.1 Interpretation

Our results *[support / partially support / do not support]* the alternative hypothesis H_1 stated in Section 3. *[Explain what the statistical result means in plain language, and connect it back to the research question from Section 1.]* This is broadly *[consistent / inconsistent]* with the findings of Smith and Doe [2020], though our effect size is *[larger / smaller / comparable]*, which may be explained by *[differences in dataset, population, or method]*.

5.2 Limitations and Threats to Validity

- **Internal validity:** *[e.g., possible confounding variables, lack of randomization, measurement error.]*
- **External validity:** *[e.g., how well the sample generalizes to other populations, time periods, or settings.]*
- **Construct validity:** *[e.g., whether the measured variables truly capture the concepts of interest.]*
- **Statistical conclusion validity:** *[e.g., small sample size, violated test assumptions, multiple-comparison risk.]*

5.3 Implications

[State the practical or theoretical implications of your findings— who should care about this result, and what should they do differently as a result?]

6 Conclusion

In this project, we investigated *[restate the research question in one sentence]* using *[briefly restate method: dataset + statistical test/design]*. Our analysis found that *[state the main finding in one or two sentences, consistent with Section 4 and Section 5]*.

These findings contribute to *[the broader area, e.g., software analytics, empirical software engineering, applied statistics practice]* by *[state the specific contribution — a new finding, a replication, a methodological point]*.

Future work could address the limitations noted in Section 5 by *[e.g., collecting a larger sample, replicating the study on a different population, using a more robust experimental design, or controlling for additional confounding variables]*.

References

- Alice Jones and Robert Brown. An example conference paper on experimental design. In *Proceedings of the Example International Conference*, pages 100–110, 2019.
- Sung-Min Lee. An example study using an observational method. *Journal of Applied Statistics*, 48(7):1200–1215, 2021.
- John Smith and Jane Doe. An example study on statistical methods in software engineering. *Journal of Empirical Software Engineering*, 25(3):1–20, 2020.

A Examples

This appendix demonstrates the conventions used throughout this template. Copy whichever snippet you need into your own section files, then delete this appendix from your final report.

A.1 Figures

Place image files in `tikz/` and reference them as follows:

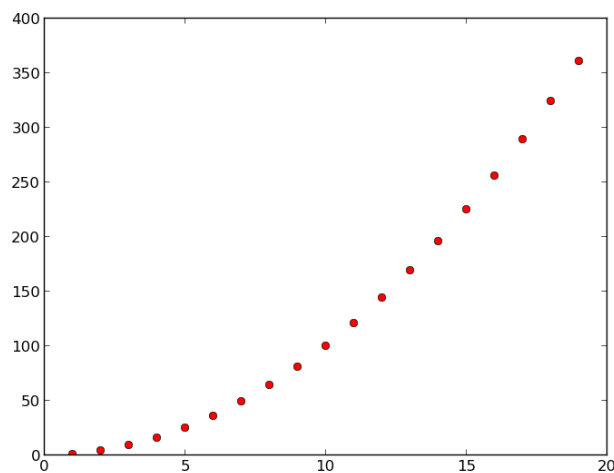


Figure 2: Example figure. Replace `tikz/example-plot.png` with your own file. Refer to it in text with `\Cref{F:example}`.

As Figure 2 shows, `cleveref` automatically inserts the word "Figure" for you — no need to type it manually.

A.2 Tables

For simple tables, use `booktabs` for clean horizontal rules (avoid vertical rules):

Table 1: Example results table.

Model	Accuracy (%)	F1-score
Baseline	72.3	0.68
Proposed	81.5	0.77

Table 1 summarizes the comparison between the baseline and the proposed approach. For larger tables, save the `tabular` environment in its own file under `tables/` and pull it in with `\input{tables/your-table}`.

A.3 TikZ figures

Students are encouraged to draw diagrams (flowcharts, experimental designs, architecture diagrams, decision trees, etc.) natively in $\text{\LaTeX}$ using `TikZ` rather than pasting in screenshots — this keeps vector-quality, theme-consistent figures that scale cleanly in the PDF.

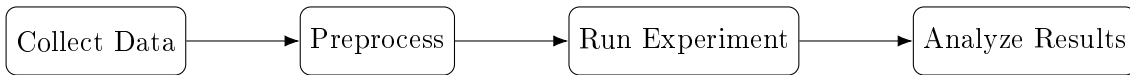


Figure 3: Example TikZ diagram of an experimental pipeline.

For more complex diagrams, consider keeping the `tikzpicture` in its own file under `tikz/` and `\input{}`-ing it, so `main.tex` stays readable.

A.4 Citations

This template uses `natbib`. Given a BibTeX entry with key `smith2020`, in `references.bib`:

- `\citep{smith2020}` renders as a parenthetical citation, e.g. [Smith and Doe, 2020] — use this when the citation is not the grammatical subject of the sentence.
- `\citet{smith2020}` renders as an in-text citation, e.g. Smith and Doe [2020] found that... — use this when you are directly discussing the authors' work.

Note: `smith2020` is a placeholder key. Add real entries to `references.bib` before compiling, or these citations will show up as "?".